React Handbook — Interview Edition

How to use this edition

The night-before companion to the full *React Handbook* (React 19.3). Every chapter is squeezed into one page or two of facts, questions and traps; nothing here is new — each line comes from the full book, where the explanation, runnable code and real output live.

Section in each chapter	Use it to…
Must-know	Recite the definitions. If one feels shaky, open the full chapter.
Key facts & numbers	Memorise the tables: defaults, versions, comparisons.
Code to remember	Write these from memory on a blank page (≤10 lines each).
In the interview	Say each answer aloud in under 30 seconds.
Gotchas	The "what does this log / why is this broken" traps.
TL;DR	The one-line summary to say first.

Plans. *One evening:* the Top 25 below, then every TL;DR and every Gotchas list. *Two evenings:* add all "In the interview" blocks and Chapter 15's eight builds. *Morning of:* the Last-minute checklist at the end. For machine-coding rounds rehearse the seven-step method (Ch 15) and rebuild at least two builds from a blank file.

Versions assumed throughout: React 19.3, React Router 8, Next.js 16, React Compiler 1.0 (Oct 2025); CRA deprecated (Feb 2025).

Top 25 questions across the book

#	Question	One-line answer	Ch
1	What happens when you call a setter?	Update queued → render (pure) → commit minimal DOM diff → paint → Effects	1, 15
2	How does reconciliation work?	O(n) diff: different type = new subtree; keys match list items; identity = type + position + key	1
3	Why keys, and why not index?	Keys carry state/DOM with the item; index keys misattach state on reorder/insert; never random	2
4	Why does `{count && <X/>}` show 0?	`&&` returns the falsy `0`, which renders; use `count > 0 &&`	2
5	Why do three `setN(n+1)` add 1?	State is a snapshot; use `setN(p => p + 1)`	3
6	What is batching?	All updates in one event → one render; automatic everywhere since React 18	3
7	Why immutable updates?	React compares by reference (`Object.is`); mutation skips renders and breaks memo	3
8	Derived state?	Compute from props/state in render; don't store or sync with Effects	3
9	How to reset a component's state?	Change its `key`	3
10	Controlled vs uncontrolled?	React state owns value vs DOM owns it; React 19 Actions favour uncontrolled	4
11	`useActionState` / `useFormStatus` / `useOptimistic`?	Last action result `(prev, formData)`; parent form pending (child, `react-dom`); temporary value with auto rollback	4
12	What is `useEffect` for?	Syncing with external systems; cleanup before re-run and on unmount	5
13	Why does my Effect run twice?	Dev-only StrictMode setup → cleanup → setup to expose missing cleanups	5
14	Fetch race conditions?	`AbortController` or ignore flag in cleanup; better TanStack Query/loaders	5
15	`useRef` vs `useState`?	Ref persists without re-rendering; state re-renders	5
16	Rules of Hooks — why?	Hook state is matched by call order; top level, React functions only	7
17	Custom Hooks share state?	No — they share logic; each call has its own state	7
18	Is Context a state manager?	No, a transport; all consumers re-render on a new value, through `memo`	6
19	How to choose state management?	Server → query cache; shareable → URL; local → hooks; rare global → context; busy global → Zustand/RTK	6
20	`memo` / `useMemo` / `useCallback` / Compiler?	Skip on shallow-equal props / cache value / cache function; Compiler auto-memoizes at build time	10
21	`useTransition` vs `useDeferredValue`?	Wrap an update you own vs defer a value you receive; neither is a debounce	7
22	Suspense and `use()`?	Fallback while lazy/`use(promise)` pending; promise must be cached; transitions keep old UI	9

#	Question	One-line answer	Ch
23	Hydration and mismatches?	Browser adopts server HTML; first client render must match; switch after mount	13
24	Server vs Client Components?	Server: async, zero JS, no state; `'use client'` marks the boundary; `'use server'` = Server Functions	13
25	What don't error boundaries catch?	Event handlers, async callbacks, SSR, their own errors	14

1. Why React, and Setting Up

What React is, declarative UI, render/commit and reconciliation, rendering strategies, project setup and JSX.

Must-know

- **React** — a UI *library* (not a framework): components (functions returning JSX) + `react-dom` to render them. Routing, data and build come from the ecosystem or a framework.
- **Declarative** — describe what the UI looks like for the current state (UI = f(state)); React works out the DOM steps. Imperative = list the steps yourself.
- **Re-render** — React calls your component function again; the DOM changes only where output differs.
- **Render vs commit** — render = call components to compute the new element tree (pure, may be repeated or discarded); commit = mutate the DOM, attach refs, run layout effects.
- **Reconciliation** — O(n) diff using two heuristics: different element types produce different trees; keys identify list items across renders.
- **Hydration** — React in the browser adopting server-generated HTML: reuses the DOM, attaches handlers and state.
- **Hydration mismatch** — server and client render different output (`Date.now()`, `Math.random()`, `window` checks during render).
- **Component** — capitalised function returning JSX, defined at **module top level**, used as `<Name />`.
- **JSX** — compiles to `jsx('h1', {className:'x', children:'Hi'})` calls that return plain immutable element objects. Automatic runtime (React 17+) means no `import React` needed.
- **Fragment** — `<>…</>` groups children without an extra DOM node; use `<Fragment key={id}>` when a list needs a key.

Key facts & numbers

Strategy	First content	Navigation	SEO	Server needed	Typical use
MPA	Fast	Full reload	Good	Yes	Wikipedia, classic admin
SPA / CSR	Slow (wait for JS)	Instant	Weaker	No (static)	Dashboards behind login
SSR	Fast	App-like after hydration	Good	Yes (Node/edge)	E-commerce, news
SSG	Fastest (CDN)	App-like after hydration	Good	No	Docs, blogs, marketing

JSX rule	Why
One root element (or `<>`)	A function returns one value
Close every tag (`<img />`)	JSX is stricter than HTML
camelCase attrs: `className`, `htmlFor`, `onClick` (`aria-*` / `data-*` keep dashes)	Attributes become JS object keys; `class` / `for` are reserved
`{}` holds expressions only (no `if` / `for`)	Use ternary, `&&`, `.map()` or a variable above `return`
`style={{ fontSize: 14 }}`	Style is an object; numbers get `px`

- React runtime ≈ 60–70 KB gzipped. CRA deprecated **February 2025**.
- New app: a framework (Next.js / React Router / Expo); SPA or learning: `npm create vite@latest app -- --template react`.
- Entry point: `createRoot(document.getElementById('root')).render(<StrictMode><App/></StrictMode>)` — create the root once.
- Renders nothing: `null`, `undefined`, `true`, `false`. Renders: `0` and `NaN`. Objects throw "Objects are not valid as a React child".

In the interview

Q: What does "React is declarative" mean? **A:** I describe the UI for the current state; on change React re-runs the component, diffs the new output against the old and applies only the differences — removing the class of bugs where part of the screen isn't updated.

Q: Explain the update cycle. **A:** Trigger → render (call components, build element tree) → commit (apply minimal DOM diff). Same type at same position = keep and update; different type = replace; lists matched by key.

Q: Why can changing a wrapper from `<div>` to `<section>` lose input text? **A:** A different type at the same position destroys and recreates the subtree, including its state.

Q: Does the virtual DOM make React faster than hand-written DOM code? **A:** Not necessarily — hand-tuned DOM code does less work. The benefit is good-enough performance automatically while writing declarative code.

Q: SPA vs SSR vs SSG? **A:** SPA ships an empty shell and renders in the browser (simple hosting, slow first paint, weaker SEO); SSR renders HTML per request then hydrates; SSG renders at build time. Frameworks choose per route.

Q: Why Vite over CRA? **A:** The dev server serves native ES modules and compiles only what the browser requests, so startup and HMR stay fast; production is still fully bundled. HMR = swap the changed module in the running page, keeping state.

Q: Is JSX safe from XSS? **A:** Values in braces are escaped. Risks: `dangerouslySetInnerHTML`, user-supplied `href` URLs (React 19 blocks `javascript:` URLs, but validate anyway) and third-party HTML.

Gotchas

- Defining a component inside another → new type every render → state reset, inputs lose focus. Move it to module level.
- Calling a component as a function `{Profile()}` → no identity in the tree, Hook rules broken. Use `<Profile />`.
- Lowercase component name renders an unknown HTML tag. Capitalise.
- `return` alone on its line returns `undefined` — start `(` on the return line.
- `src="{avatar}"` is the literal string; use `src={avatar}`.
- SSR makes content *visible* sooner, not *interactive* sooner (that waits for JS + hydration).

- Don't remove `<StrictMode>` to "fix" doubled logs — the doubles reveal bugs.

TL;DR

React = components + React DOM; trigger → render → commit with a keyed, type-based diff. Pick CSR/SSR/SSG per route; start with a framework or Vite.

2. Components and Props

Function components, props, `children`, composition, purity, conditional rendering, lists and keys.

Must-know

- **Component** — capitalised function `(props) => JSX`; may return JSX, string/number, array or `null`. Only host elements it returns reach the DOM.
- **Props** — one **read-only** object from the parent; one-way data flow: data down via props, events up via callback props.
- **Defaults** — destructuring defaults `({ size = 100 })`. `defaultProps` is **ignored** on function components in React 19.
- **`children`** — nested JSX arrives as `props.children`; **slots** = JSX passed in named props (`icon={<Star />}`).
- **Composition** — nesting components instead of inheritance; the first cure for prop drilling (top component creates the deep child and passes it through as children).
- **Purity** — same props/state/context → same JSX; no mutation of anything that existed before render. Local mutation of values created in this render is fine.
- **StrictMode** — double-renders in development to expose impure components.
- **Presentational vs feature** — presentational take props and render UI; feature components know the data and wire things together.
- **Key** — stable, sibling-unique identity used to match list items across renders; consumed by React (not readable as `props.key`).

Key facts & numbers

Condition pattern	Use for
`if` / early `return`	Large branches
`cond ? <A/> : <B/>`	Either-or
`cond && <A/>`	Show or nothing — **boolean on the left**
`return null`	Render nothing

Key choice	Verdict
Item ID from data / UUID created with the item	Correct
Array index	Only for static lists (never reordered/filtered/prepended, no per-item state)
`Math.random()` / `crypto.randomUUID()` in JSX	Never — remounts every item each render

Spread `{...props}`	List props explicitly
Thin wrappers around native elements (`Button`, `Input`); later props override earlier	Feature components — clear API

Code to remember

```
function Button({ variant = 'primary', ...rest }) {
  return <button className={`btn ${variant}`} {...rest} />;
}
const list = items.map(item => <Row key={item.id} id={item.id} {...item} />);
```

In the interview

Q: What are props and why are they read-only? **A:** Read-only inputs from the parent. The parent owns the data; if a child mutated it, siblings would show stale values and React couldn't know to re-render.

Q: Why do keys matter? **A:** React matches items between renders by key, so it can move DOM nodes and keep state with the right item. Index keys break on reorder/filter/prepend — state stays at the position, not with the item.

Q: Do keys need to be globally unique? **A:** No — only among siblings.

Q: Why does `{count && <Badge />}` show "0"? **A:** `&&` returns its falsy left operand and 0 is renderable. Use `count > 0 &&`.

Q: Composition vs Context? **A:** Composition (children/slots) first; Context when many components at different depths need the same value (theme, auth, locale).

Q: Why must components be pure? **A:** React may render several times or discard a render; side effects go in event handlers, or Effects for syncing external systems.

Q: Can a prop be a component? **A:** Yes — an element (`icon={<Star />}`) or a type (`as={Link}`) rendered as `<Icon />` after capitalising.

Gotchas

- `size="50"` is a string → `size={50}`.
- `function Avatar(person, size)` → forgot braces; props arrive as one object.
- Copying a prop into state ignores later parent updates.
- `items.sort()` mutates the parent's array → `items.toSorted()` or `[...items].sort()`.
- Key on the `<li>` inside the child instead of on the element returned from `map` → warning persists.
- Multiple siblings per item → `<Fragment key={id}>` (short `<>` can't take a key).

TL;DR

Props flow down read-only; callbacks flow up. Compose with `children`. Render must be pure. `map` + stable ID keys; boolean on the left of `&&`.

3. State and Events

Event handlers, `useState`, snapshots, batching, updaters, immutable updates, lifting state, derived state, controlled inputs, state position and `key`.

Must-know

- **Handler** — function passed as an `on…` prop: **pass, don't call** (`onClick={handle}`; `() => handle(id)` for args). Handlers are *the* place for side effects.
- **SyntheticEvent** — React's wrapper mirroring the native event. `e.target` = where the event started; `e.currentTarget` = element whose handler runs.
- `stopPropagation` stops bubbling; `preventDefault` stops the browser default (form submit reload).

- **`useState`** — `[value, setValue]`; memory that persists between renders, private per instance, identified by Hook call order. Setter schedules a re-render.
- **Lazy initialiser** — `useState(fn)` calls `fn` once; `useState(fn())` runs every render.
- **Snapshot** — state is a constant within one render; setters never change the variable you hold.
- **Batching** — all setter calls in one event → one re-render (everywhere since React 18); `flushSync` opts out.
- **Updater function** — `setX(prev => next)` builds on the queued value; plain `setX(v)` replaces.
- **Immutability** — React compares by reference (`Object.is`); mutating and setting the same object may not re-render.
- **Lifting state up** — move shared state to the closest common parent; pass value + callback down. "As low as possible, as high as necessary."
- **Derived state** — compute from props/state in render; don't store. Store IDs, not copies; one `status` over contradictory booleans.
- **Controlled input** — `value={state}` + `onChange`; checkbox uses `checked` / `e.target.checked`. Initialise with `''`.
- **State position** — same type + same position = state kept; different type or different `key` = reset. Use `key` to reset on purpose.

Key facts & numbers

Value	Where it belongs
Never changes	Constant outside the component
From the parent	Props (don't copy into state)
Computable from props/state	Variable in render
Persists but must not re-render (timer ID, DOM node)	`useRef`
Shared by distant components	Lift up → context/store
Server data	TanStack Query

Immutable update	Code
Object field	`{ ...obj, key: v }`
Add / remove	`[...arr, x]`, `arr.filter(i => i.id !== id)`
Replace one	`arr.map(i => i.id === id ? {...i, done: true} : i)` or `arr.with(idx, v)`
Sort / insert	`toSorted()`, `toSpliced()`
Deep	Copy every level, or Immer (structural sharing)

Controlled	Uncontrolled
Instant validation, formatting, dependent fields; re-render per keystroke	Less code; DOM holds value; read on submit via `FormData`/refs; first-class with React 19 Actions

Code to remember

```
const [n, setN] = useState(0);
function addThree() {           // from 0 → 3; three setN(n + 1) → 1
  setN(p => p + 1); setN(p => p + 1); setN(p => p + 1);
}
// setN(n + 5); setN(m => m + 1)  from 0 → 6
// setN(m => m + 1); setN(n + 5)  from 0 → 5 (last replace wins)
```

In the interview

Q: Why not a local variable instead of state? **A:** Locals reset every render and changing them doesn't trigger one.

Q: Why do three `setCount(count + 1)` calls add 1? **A:** They all read the same snapshot; use the updater form to chain.

Q: Why must state be immutable? **A:** React detects change by reference; mutation may skip the re-render and breaks snapshots and memoisation.

Q: Is `[...arr]` a deep copy? **A:** No — shallow; elements are shared.

Q: Should an Effect update derived state? **A:** No — extra render and a flash of stale UI. Compute during render (`useMemo` if expensive).

Q: How do you reset a component's state when a prop changes? **A:** Give it a `key` that changes (e.g. `key={userId}`); different props alone don't reset state.

Q: When do you lift state? **A:** When siblings need the same data — closest common parent owns it; children become controlled. If it goes too high, use composition or context.

Gotchas

- `onClick={handle()}` calls it during render → pass the function.
- Clickable `<div>` → use `<button>` (keyboard + screen reader).
- Forgetting `preventDefault` in `onSubmit` → page reload, state lost.
- `arr.push(x); setArr(arr)` → no re-render.
- `setCount(count++)` → passes the old value.
- Side effects inside an updater → may run twice.
- Stale closure: a timer created once reads an old snapshot → updater or ref.
- `value` without `onChange` → read-only + warning; `value={undefined}` then string → "uncontrolled to controlled" warning.
- `selected` on `<option>` → put `value` on `<select>`.

TL;DR

State = per-instance snapshot; setters queue a batched re-render. Use updaters to chain, never mutate, keep minimal state and derive the rest, and use `key` to reset.

4. Forms and Actions

Form submission, controlled vs uncontrolled, validation, React 19 form Actions, `useActionState`, `useFormStatus`, `useOptimistic`, Server Functions, React Hook Form + Zod.

Must-know

- **`FormData`** — reads fields by **`name`**; unchecked boxes are missing; all values are strings (`Number(fd.get('age'))`); `getAll` for repeated names.
- **Controlled vs uncontrolled** — React state owns the value (`value` + `onChange`, re-render per keystroke) vs the DOM owns it (`defaultValue`, read via `FormData`/ref). File inputs are always uncontrolled.
- **Action** (React 19) — a function run inside a **Transition**, e.g. passed to `action`/`formAction` or called in `startTransition`.
- **Form Action** — `<form action={fn}>`: React prevents navigation, calls `fn(formData)` in a Transition (async allowed), **auto-resets uncontrolled fields after success**; `formAction` on a button overrides per button; uncaught errors → error boundary.

- **`useActionState`** — `const [state, action, isPending] = useActionState(async (prev, formData) => next, initial)`; the natural place to return validation errors or success.
- **`useFormStatus`** — from **`react-dom`**, called in a **child** of the form: `{ pending, data, method, action }`. Ideal for a reusable Submit button.
- **`useOptimistic`** — `useOptimistic(realValue, reducer)` shows the expected result while an Action runs; when it ends React drops the optimistic value (success → real data takes over; failure → automatic rollback).
- **Server Function** — `'use server'` function run on the server but called like a local function; as a form action it enables **progressive enhancement** (form works before JS loads).
- **React Hook Form + Zod** — RHF manages big forms with uncontrolled inputs; Zod defines rules once for client and server; `zodResolver` connects them.

Key facts & numbers

	`onSubmit` + `preventDefault`	**`action={fn}`**
Version	Any	React 19+
Input	Full event	`FormData` only
Pending / reset / optimistic / progressive enhancement	Manual	Built in
Fits	Controlled fields	Uncontrolled fields

When to show errors	**Use for**
On every change	Only positive feedback (password strength)
On blur (touched)	Most fields
On submit only	Short forms (login)
On blur, then live while fixing	Default recommendation

`useActionState` rule	**Detail**
Signature	Action receives **`(prevState, payload)`** — prev first
Queue	Dispatches run **sequentially**, not in parallel
Reset	No built-in reset: return `initialState` or remount with `key`
Errors	Throwing cancels queued actions and shows the error boundary → return errors
Manual dispatch	Wrap in `startTransition`
Server Functions	`initialState` and payload must be serializable

Code to remember

```
function Submit() {                       // child of <form>
  const { pending } = useFormStatus();    // from 'react-dom'
  return <button disabled={pending}>{pending ? 'Saving…' : 'Save'}</button>;
}
const [state, formAction, isPending] = useActionState(async (prev, fd) => {
  const email = fd.get('email');
  if (!email.includes('@')) return { error: 'Invalid email', email };
  await save(email); return { ok: true };
}, { error: null });
// <form action={formAction}><input name="email" defaultValue={state.email} /><Submit/></form>
```

In the interview

Q: Controlled or uncontrolled in React 19? **A:** Default to uncontrolled fields with a form Action; use controlled where the UI must react while typing (formatting, counters, dependent fields).

Q: How do you validate a form? **A:** Layered: HTML constraints, a pure `validate()` for custom/cross-field rules, and **always** server validation. Derive errors, show after blur/submit, `aria-invalid` + `aria-describedby`, focus the first invalid field; Zod to share rules.

Q: What does `<form action={fn}>` do? **A:** Prevents navigation, passes `FormData`, runs in a Transition so pending state is visible to `useFormStatus`/`useActionState`, and resets uncontrolled fields after success.

Q: How does `useOptimistic` roll back? **A:** The optimistic value lives only for the Action's duration; if the real state wasn't updated, the old value shows again.

Q: How do you keep user input after a failed server validation with uncontrolled fields? **A:** Return the submitted values from the action and use them as `defaultValue`s.

Q: What must you do inside a Server Function? **A:** Treat it as a public API endpoint: authenticate, authorise and validate; arguments/returns must be serializable.

Q: How do you reset an uncontrolled form? **A:** `form.reset()` (a successful Action does it automatically) or change the form's `key`.

Gotchas

- Writing `async (formData) => …` for `useActionState` → you get the previous state as "formData".
- `useFormStatus` in the same component as `<form>` → always `pending: false`; import from `react-dom`, not `react`.
- Throwing expected errors ("email taken") shows the error boundary → return them.
- Missing `name` attributes → `FormData` silently empty.
- Changing `defaultValue` doesn't update the field → remount with `key` or go controlled.
- `useOptimistic` setter must be called inside an Action; its reducer must be pure; show a "sending" marker.
- Don't use optimistic UI for payments/bookings or unpredictable server results.
- Disabling Submit until valid hides *why* — allow submit and show errors.
- `'use server'` marks server-callable functions, not Server Components.

TL;DR

React 19 forms: `<form action>` + `useActionState` (prev, formData) + `useFormStatus` in a child + `useOptimistic`; return errors, validate on the server, prefer uncontrolled fields.

5. Effects and Refs

Where side effects belong, `useEffect` and cleanup, StrictMode's double run, dependencies, "you might not need an Effect", fetch races, `useRef`, DOM refs, `ref` as a prop, `useLayoutEffect`.

Must-know

- **Where code goes** — pure calculation → render; caused by a user action → event handler; needed because the component is on screen, to sync with something outside React → **Effect**.
- **`useEffect`** — synchronises with an external system. Setup runs **after commit** (normally after paint); cleanup runs before the next setup and on unmount. Think start-sync / stop-sync, not lifecycle.
- **Cleanup** — a function that exactly undoes setup: disconnect, clear timer, remove listener, abort. It sees the *old* render's values.

- **StrictMode double run** — dev-only setup → cleanup → setup on mount; exposes missing cleanups. Fix the cleanup; never suppress.
- **Dependencies** — every reactive value the Effect reads, compared with `Object.is`. Never lie to the linter; reduce deps by changing code.
- `useEffectEvent` (stable since 19.2) — non-reactive logic read from an Effect ("connected, show a toast with the current theme"). Call only from Effects; don't list in deps; don't pass to other components.
- `useRef` — stable `{ current }` box that persists; mutating it **doesn't re-render**. For timer IDs, DOM nodes, third-party instances. Don't read/write during render.
- **DOM ref** — `ref={r}` → `r.current` is the node after commit (`null` during first render). Lists: ref callback + `Map`; React 19 ref callbacks may **return a cleanup**.
- `ref` **as a prop** (React 19) — `function Field({ ref }) { return <input ref={ref} /> }`; `forwardRef` is legacy. `useImperativeHandle` exposes a limited API (e.g. `focus`). React 19.3 adds refs on `<Fragment>`.
- `useLayoutEffect` — runs after DOM update, **before paint**; for measure-then-adjust without flicker. Blocks painting; does nothing on the server.

Key facts & numbers

Dependency array	Runs
none	After every render
`[]`	On mount only (plus dev double run)
`[a, b]`	When `a` or `b` changes (`Object.is`)

Reduce deps by…	Example
Move object/function inside the Effect	Build `options` inside setup
Updater function	`setCount(c => c + 1)` removes `count`
Split Effects	One Effect per sync concern
`useEffectEvent`	Read latest non-reactive values

You might not need an Effect	Instead
Derived data (filtered list, total)	Compute in render (`useMemo` if expensive)
Reset all state on prop change	`key`
Logic caused by an event (POST on submit)	Event handler
Notifying the parent	Call its callback in the same handler
Chains of Effects setting state	Compute everything in one handler
Subscribing to an external store	`useSyncExternalStore`

Code to remember

```
useEffect(() => {
  const ctrl = new AbortController();
  setStatus('loading');
  fetch(`/api/cities?q=${q}`, { signal: ctrl.signal })
    .then(r => { if (!r.ok) throw new Error(r.status); return r.json(); })
    .then(data => { setCities(data); setStatus('done'); })
    .catch(e => { if (e.name !== 'AbortError') setStatus('error'); });
  return () => ctrl.abort();          // stale response can't overwrite a newer one
}, [q]);
```

In the interview

Q: Explain `useEffect`. **A:** Syncs with an external system after commit; cleanup before the next run and on unmount; deps decide when it re-runs.

Q: Why can't the setup be `async`? **A:** It must return nothing or a cleanup, not a promise — define an async function inside and call it.

Q: Why does my Effect run twice in development? **A:** StrictMode mounts, cleans up and mounts again to verify the cleanup mirrors setup; not in production.

Q: How do you avoid fetch race conditions? **A:** An `ignore` flag set in cleanup or an `AbortController` aborted in cleanup; plus loading/error states and `response.ok`. In production prefer TanStack Query or loaders (caching, dedup, retries, no waterfalls).

Q: `useRef` vs `useState`? **A:** Ref persists without re-rendering — for non-visual values; anything on screen belongs in state.

Q: `useEffect` vs `useLayoutEffect`? **A:** `useEffect` is the default and doesn't block paint; `useLayoutEffect` runs before paint to measure and reposition without flicker.

Q: Where does "POST analytics on submit" go vs "log page view"? **A:** Submit → handler; page view → Effect.

Gotchas

- `removeEventListener('x', () => fn())` removes nothing → keep the function in a variable.
- Cleanup for something that can't be undone (one-off POST) → belongs in a handler.
- Showing "AbortError" after cancellation → filter it out.
- Storing fetched data *and* a filtered copy → store data, derive the filter.
- The online `usePrevious` (ref written in Effect, read in render) breaks the ref rule → store previous value in state.
- Editing DOM that React manages via refs → conflicts; use refs for focus, scroll, measurement, third-party widgets.

TL;DR

Effects sync with the outside world and must clean up; most "Effects" should be render calculations, `key` resets or handler code. Refs are non-rendering boxes and DOM handles.

6. State Management

Kinds of state, `useReducer`, Context and its performance, reducer + context, Redux Toolkit, Zustand, TanStack Query, choosing.

Must-know

- **Kinds of state** — local, complex local, shared client, server, URL, form, derived. Each has a home; keep state as local as possible; never store what you can derive.
- **`useReducer`** — `const [state, dispatch] = useReducer(reducer, initial, init?)`. Reducer = **pure** `(state, action) => next`; `dispatch` is **stable**; third arg is a once-only initialiser.
- **Actions describe events** — `{ type: 'task_completed', id }`, not setters. One user action = one dispatch. No fetch, `Date.now()` or random IDs in reducers (StrictMode runs them twice).
- **Context** — `createContext(default)` → `<Ctx value={v}>` (older: `.Provider`) → `useContext(Ctx)` or `use(Ctx)`. Nearest provider wins; default used when none.
- **`use(Context)`** — can be called conditionally (after early returns, in loops); `useContext` follows Hook rules.
- **Context performance** — all consumers re-render when the value's **identity** changes; no selectors. Memoise the value, split state from dispatch, split by concern.
- **Reducer + context** — provider with `useReducer` + separate state and dispatch contexts + `useX()` Hooks that throw outside the provider = dependency-free store.
- **Redux Toolkit** — `createSlice` (reducers + action creators, **Immer** so `state.count++` is safe), `configureStore` (thunks, DevTools, checks), `useSelector` (re-renders only when the selected value changes), `useDispatch`, `createAsyncThunk`, `createSelector`, **RTK Query** for server data.
- **Zustand** — `const useStore = create((set, get) => ({ ...state, ...actions }))`; select `useStore(s => s.count)`; `useShallow` for multiple values; **no provider**; `getState/setState/subscribe` outside React; `persist`/`devtools` middleware.
- **TanStack Query** — server-state cache: `useQuery({ queryKey: [...vars], queryFn })`, `useMutation` + `invalidateQueries`. The key must include every variable.

Key facts & numbers

State kind	Home
Server data	Query cache (TanStack Query / RTK Query)
Shareable UI (filters, page)	URL
Local	`useState` / `useReducer`
Rarely-changing app-wide (theme, locale, auth)	Context
Frequently-changing shared client	Zustand (simple) / Redux Toolkit (structure, tooling)

TanStack Query	Default / meaning
`staleTime`	**0** — how long data counts as fresh
`gcTime`	**5 min** — how long unused data stays in memory
Retries	**3**
Refetch triggers	Window focus, reconnect, mount (when stale)
`isPending` vs `isFetching`	No data yet vs any request in flight (incl. background)

Redux Toolkit	Zustand
Structured, conventions for big teams, great DevTools (time travel)	Tiny API, almost no boilerplate
Needs `<Provider>`	No provider; module singleton (store-per-provider for SSR/tests)
More files/concepts; overkill for small apps	Fewer conventions; easy to grow a giant store

Code to remember

```
function reducer(state, action) {
  switch (action.type) {
    case 'added':   return [...state, { id: action.id, text: action.text, done: false }];
    case 'toggled': return state.map(t => t.id === action.id ? { ...t, done: !t.done } : t);
    case 'deleted': return state.filter(t => t.id !== action.id);
    default: throw new Error('Unknown action ' + action.type);
  }
}
```

In the interview

Q: When `useReducer` over `useState`? **A:** Many handlers update the same state or transitions are complex; the reducer centralises logic and is unit-testable without React.

Q: Can a reducer fetch data? **A:** No — pure. Fetch in a handler/Effect and dispatch the result.

Q: Why is Context not a state manager? **A:** It only delivers a value; every consumer re-renders when the value changes; no selectors or middleware.

Q: Why might a Redux component re-render on every dispatch? **A:** Its selector returns a new object/array each call; use `createSelector` or select primitives.

Q: Redux vs Zustand? **A:** Both selector-based. RTK for large teams wanting structure, DevTools and RTK Query; Zustand when that is overkill.

Q: What is server state and why a library? **A:** A cache of data someone else owns — needs fetching, caching, dedup, refetching, invalidation. With TanStack Query most "global state" disappears.

Q: `staleTime` vs `gcTime`? **A:** Freshness (stale triggers refetch) vs how long unused cache entries live.

Q: How do you choose state management? **A:** Classify: server → query cache; shareable → URL; local → hooks; rare global → context; busy global → Zustand/RTK.

Gotchas

- `value={{ user, setUser }}` inline in a provider → new object each render → all consumers re-render; memoise or split.
- Query key `['user']` for every id → users overwrite each other's cache.
- Copying query `data` into state/Redux → lose updates; derive in render.
- `new QueryClient()` in a component body → new cache each render; create once.
- Non-serializable data in Redux breaks DevTools/persistence/SSR.
- Form input text rarely belongs in Redux.
- Reading `state` right after `dispatch` → old snapshot.

TL;DR

Classify state first. Reducers are pure and event-based; Context is for rarely-changing values; stores add selectors; server data belongs in a query cache.

7. Custom Hooks and the Rules of Hooks

How Hooks work, the two rules, custom Hooks (`useLocalStorage`, `useDebounce`, `useFetch`), `useEffectEvent`, `useId`, `useSyncExternalStore`, `useTransition`, `useDeferredValue`.

Must-know

- **How Hooks work** — React stores Hook state in a per-component list matched by **call order**; change the count/order and values land in the wrong slots ("Rendered more/fewer hooks").
- **Rules of Hooks** — (1) call only at the **top level** (no `if`, loops, after early return, in `try`, in handlers); (2) only from components and custom Hooks. Enforced by `eslint-plugin-react-hooks`.
- `use` — may be called conditionally because it stores nothing in a Hook slot (reads a promise or context).
- **Custom Hook** — function named `useX` that calls other Hooks; shares **logic, not state** (each call independent). The `use` prefix lets the linter check it.
- `useEffectEvent` (19.2) — latest callback inside an Effect without re-subscribing; older code used the "latest ref" pattern.
- `useId` — SSR-safe unique ID per component for `htmlFor` / `aria-*`; **never for keys**.
- `useSyncExternalStore(subscribe, getSnapshot, getServerSnapshot)` — official way to read mutable outside data; prevents **tearing** (parts of UI showing different store versions in one render). Zustand and React Redux use it.
- `useTransition` — `[isPending, startTransition]`; marks updates non-urgent, interruptible by urgent input; old UI stays while pending. Async callback = an **Action** (React 19).
- `useDeferredValue(v)` — urgent render with the old value, background render with the new one; pair with `memo`; `initialValue` in 19.

Key facts & numbers

Hook	Wraps	Use when
`useTransition`	The **update** (you own the setter)	Tab switch, filtering, navigation
`useDeferredValue`	A **value** (e.g. received as a prop)	Slow list behind fast input
Debounce	Time	Reduce network requests

`useLocalStorage` concern	Note
SSR	No `localStorage` on the server → hydration mismatch; render default then load in Effect, or `useSyncExternalStore` + `getServerSnapshot`
Same key, same tab	Instances don't see each other (`storage` event fires only in *other* tabs)
Size	~5 MB per origin, strings only, synchronous
Security	Readable by any script — no tokens or personal data

Code to remember

```
function useDebounce(value, delay = 300) {
  const [debounced, setDebounced] = useState(value);
  useEffect(() => {
    const id = setTimeout(() => setDebounced(value), delay);
    return () => clearTimeout(id);
  }, [value, delay]);
  return debounced;
}
```

```
function useInterval(callback, ms) {
  const onTick = useEffectEvent(callback);
  useEffect(() => { const id = setInterval(() => onTick(), ms); return () => clearInterval(id); },
[ms]);
}
```

In the interview

Q: State the Rules of Hooks and why. **A:** Top level only, React functions only — order must be identical every render because state is matched by position.

Q: How do you conditionally run an Effect? **A:** Always call `useEffect`; put the `if` inside and include the condition in deps.

Q: Other causes of "Invalid hook call"? **A:** Calling a Hook outside a component, or two copies of React in the bundle.

Q: Do two components using a custom Hook share state? **A:** No — each call has its own state; share via lifting, context or a store.

Q: Custom Hooks vs HOCs / render props? **A:** Hooks compose without extra tree nesting and make data flow explicit; the others are rarely needed now.

Q: Write `useFetch`. **A:** One `{status, data, error}` state, fetch in an Effect keyed on URL, AbortController aborted in cleanup, check `res.ok` (fetch doesn't reject on 404); prefer TanStack Query in production.

Q: What if `getSnapshot` returns a new array each call? **A:** Infinite re-render loop; React warns it "should be cached".

Q: Is `useTransition` a debounce? **A:** No timer — it runs ASAP at lower priority and yields to urgent work; it can't control a text input's value.

Gotchas

- Early `return <Spinner/>` above a Hook → move all Hooks above the first return.
- Hook in `.map()` → extract an `<Item>` component.
- Silencing `exhaustive-deps` almost always hides a stale-value bug.
- Naming a Hook `getUser` → linter can't check; use `useUser`.
- Debounce without cleanup → every keystroke fires; debouncing the input's own value → caret lag; `debounce(fn)` inline in render → new timer each render.
- `useDeferredValue` without `memo` on the slow child gains nothing; deferring a new object each render defeats it; it doesn't reduce requests.
- External store updates can't be transitions → React falls back to a blocking render.

TL;DR

Hooks are matched by call order, so call them unconditionally. Custom Hooks share logic; `useSyncExternalStore` for outside data; transitions/deferred values keep input responsive but aren't debounces.

8. Styling React Apps

Plain CSS, inline styles, CSS Modules, class composition, Tailwind v4, CSS-in-JS, theming and dark mode, component libraries.

Must-know

- **Plain CSS** — `className` + imported CSS file; imports are **global**, so scope with a naming convention (BEM) or `@layer`. Modern CSS has native nesting, cascade layers and custom properties.
- **Inline `style`** — camelCase object; numbers → `px` except **unitless** props (`lineHeight`, `zIndex`, `opacity`, `flexGrow`); custom properties (`'--accent'`) pass through; **no** `:hover`, pseudo-elements, media queries or keyframes. Best for dynamic values, ideally via CSS variables.
- **CSS Modules** — `Card.module.css` + `styles.card`; class names renamed to be local; real CSS, zero runtime, works with Server Components; `composes`, `:global`.
- **Class composition** — `clsx`/`cx`; accept a `className` prop; variant maps instead of ternary chains.
- **Tailwind CSS v4** — utility-first classes in JSX; the build **scans source text** and generates only used CSS; theme configured in CSS with `@theme`; conflicts resolved with `tailwind-merge`.
- **CSS-in-JS** — runtime (styled-components, Emotion): convenient, dynamic from props, but render cost, bigger bundles, clash with Server Components. Zero-runtime (vanilla-extract, StyleX, Panda) output static CSS.
- **Theming** — tokens as CSS variables overridden by `[data-theme=dark]`; provider resolves light/dark/system and writes the attribute; a **head script** prevents the flash; `color-scheme` for native controls; `light-dark()`.
- **Component libraries** — styled kit (speed) vs headless (custom design + accessibility) vs copy-in (shadcn/ui, own the code). Accessibility is the reason not to hand-roll complex widgets.

Key facts & numbers

Approach	Scoped	Runtime	Works with RSC	Dynamic values
Plain CSS	No (global)	None	Yes	CSS variables
Inline style	Element	None	Yes	Direct (no hover/media)
CSS Modules	Yes	None	Yes	CSS variables / inline
Tailwind	Utility classes	None	Yes	Arbitrary values / CSS variables
Runtime CSS-in-JS	Yes	Yes	Friction	Props
Zero-runtime CSS-in-JS	Yes	None	Yes	Build-time tokens

- WCAG 2.2 contrast: **4.5:1** for normal text (dark themes too).

In the interview

Q: Compare styling approaches. **A:** Global CSS is simple but needs conventions; CSS Modules scope per file with zero runtime; Tailwind generates only used utilities; runtime CSS-in-JS adds render cost and doesn't fit Server Components; inline styles only for truly dynamic values.

Q: How do you implement dark mode? **A:** CSS-variable tokens, `[data-theme=dark]` overrides, a provider for the toggle, a head script against flash, `color-scheme` for native controls.

Q: CSS Modules vs Tailwind? **A:** Both scoped and zero-runtime; Modules keep semantic CSS in CSS files, Tailwind keeps styling in markup with a token scale. Team preference decides.

Q: Why `className`, not `class`? **A:** `class` is reserved in JavaScript, and it matches the DOM property `element.className`.

Q: Why does `` `bg-${color}-500` `` not work in Tailwind? **A:** The scanner reads source text and never sees the full class, so no CSS is generated — map to complete class names.

Gotchas

- `Card.css` without `.module` → nothing scoped, `styles` empty.

- `className="card"` with CSS Modules → no style (real name is renamed); typo `styles.tittle` → silently `undefined`.
- `px-2 px-4` → winner decided by generated CSS order, not attribute order → `tailwind-merge`.
- Tailwind Preflight resets headings/lists — intended; style explicitly.
- Hard-coded colours ignore the theme; reading `localStorage` during SSR crashes → client or cookie.
- Dark mode is not just inverted colours: adjust shadows, borders, accents.
- Mixing two styled kits; fighting a styled kit with overrides (go headless); leaking a library's API (wrap it once in your own `<Button>`).

TL;DR

Pick a zero-runtime scoped approach (CSS Modules or Tailwind), use CSS variables for themes and dynamic values, and lean on headless libraries for accessible widgets.

9. Routing and Data Fetching

Client-side routing, React Router 8 (declarative, data, framework modes), layouts, loaders, actions, fetchers, middleware, TanStack Query, mutations, Suspense and `use()`, loading UX.

Must-know

- **Client-side routing** — History API + route matching; the URL is shareable state. The server must return `index.html` for every app path (else refresh on `/products/3` 404s).
- **React Router 8** — one package: `npm i react-router`; `RouterProvider` from `react-router/dom`; middleware on by default.
- **Routes** — `path`, `:param` (`useParams`, **strings**), `?search` (`useSearchParams`), `*` for 404, `<Link>`/`<NavLink>`, `useNavigate` in code.
- **Nesting** — parent renders `<Outlet/>`; `index` route = default child; pathless routes for shared layouts; relative links; `useOutletContext`.
- **Loader** — runs **before render**, in parallel for all matched routes; `useLoaderData()`; `useNavigation()` for pending UI; `throw data(msg, {status})` handled by a route `ErrorBoundary`.
- **Action** — `<Form method="post">` → route `action` → return errors (`useActionData`) or `redirect`; loaders **revalidate automatically**; `useFetcher` for submissions that shouldn't navigate.
- **Middleware** — before/after hooks around loaders (auth guards, logging, context via `createContext` + `context.get/set`). `lazy` splits route code.
- **TanStack Query** — caches by key, dedupes, retries, background-refetches stale data on mount/focus/reconnect. `staleTime` 0, `gcTime` 5 min.
- **Query keys** — every input in the key, general → specific; `queryOptions` for reuse; `keepPreviousData` for pagination; `enabled` for dependent queries; `useInfiniteQuery` for infinite lists.
- **Optimistic mutation** — cancel → snapshot → set → rollback on error → invalidate on settle.
- **`use(promise)`** — suspends until resolved; needs `<Suspense>` + an error boundary; the promise must be **cached**, never created in render. `useSuspenseQuery` = Suspense with a cache.

Key facts & numbers

RR mode	You write	Adds	Use when
Declarative	`<BrowserRouter>` + `<Routes>`/`<Route>`	Matching, links, params	Simple SPAs, own data fetching
Data	Route objects → `createBrowserRouter` + `<RouterProvider>`	Loaders, actions, pending, error boundaries, fetchers, middleware	Data tied to navigation
Framework	Route modules + Vite plugin	SSR, per-route splitting, typegen, pre-rendering	Full-stack (Remix successor)

Loading UX	Show
First load	Skeleton
Navigation	Old content + pending indicator (transition)
Fast refresh	Nothing
Background refetch fails with cached data	Data + subtle warning
Validation error	Inline
Empty	Separate "No results" state

Code to remember

```
const router = createBrowserRouter([{ path: '/', Component: Layout, children: [
  { index: true, Component: Home },
  { path: 'products/:id', Component: Product,
    loader: ({ params }) => qc.ensureQueryData(productQuery(params.id)) },
]}]);
// Product: const { data } = useSuspenseQuery(productQuery(useParams().id));
```

In the interview

Q: Why loaders instead of fetching in `useEffect`? **A:** Loaders start at navigation, before render, in parallel — removing render-then-fetch waterfalls; pending and errors are handled by the router.

Q: Why both a loader and TanStack Query? **A:** The loader removes the waterfall; the query cache removes repeated requests and keeps data fresh (shared `queryOptions`, `ensureQueryData` in the loader, `useSuspenseQuery` in the component).

Q: Why not Redux for server data? **A:** You'd hand-write caching, dedupe, refetching and invalidation.

Q: `isPending` vs `isFetching`? **A:** No data yet vs any request in flight, including background refetch.

Q: What happens on window focus? **A:** Stale queries with mounted observers refetch.

Q: Where do filters/page number live? **A:** In search params — shareable, survives refresh.

Q: How do you avoid flashing Suspense fallbacks on navigation? **A:** Wrap the update in a transition so old content stays visible.

Gotchas

- `<a href>` inside the app → full reload; use `<Link>` (keep `<a>` for external).
- `navigate()` during render → call in handlers/Effects or render `<Navigate>`.
- Query function that doesn't throw on HTTP errors → check `res.ok`.
- `isLoading` to test for data → use `isPending` or check `data`.
- Turning off refetching because it's noisy → raise `staleTime`.

- One global spinner; `catch {}` swallowing errors → stuck in "Loading…".
- Creating a promise in render for `use()` → infinite suspend; cache it.

TL;DR

Router data mode for fetch-on-navigation, TanStack Query for caching, Suspense + error boundaries for states, URL for shareable UI state.

10. Performance

The render cycle, re-render triggers, the Profiler, `memo`, `useCallback`/`useMemo`, React Compiler, `lazy` code splitting, virtualization, common bugs, checklist.

Must-know

- **Render cycle** — trigger → render (call components) → commit (DOM diff) → effects. A re-render is a function call, not a DOM update.
- **Re-render triggers** — own state, **parent re-render**, context it reads, subscribed external store. Props are *not* compared unless `memo`. Setting state to an identical value bails out. Ref changes never re-render.
- **Children from above** — elements passed as `children` keep identity, so they're skipped when the wrapper re-renders (free optimisation).
- **Measure first** — DevTools "highlight updates" + Profiler ("why did this render?"), Performance tracks (19.2+), `<Profiler onRender>`; compare `actualDuration` vs `baseDuration`. Profile a production (profiling) build with CPU throttling.
- `memo` — skip re-render when props are shallow-equal (`Object.is`). Doesn't stop own-state or context updates. A performance **hint**, not a guarantee.
- `useCallback` / `useMemo` — cache a function / computed value until deps change. Matter only when the reference is compared (by `memo`, Effect deps, a caller). Cache calculations **≥ ~1 ms**.
- **React Compiler** — build-time Babel plugin, **v1.0 stable October 2025**; auto-memoizes values, callbacks and JSX; requires the Rules of React; `"use no memo"` escape hatch; "Memo ✨" badge in DevTools.
- **Code splitting** — `const Page = lazy(() => import('./Page'))` at **module level** + `<Suspense fallback>` + error boundary; needs a `default` export; split by route/heavy widget; preload on intent; `startTransition` to avoid fallback flashes.
- **Virtualization** — render visible rows + overscan in a full-height spacer; `first = floor(scrollTop / rowHeight) − overscan`. 10,000 items → ~16 DOM rows. Production: TanStack Virtual / react-window.

Key facts & numbers

Plain list	Virtualized
DOM grows with data	Constant (~visible rows)
Ctrl+F finds any row	Only rendered rows
Screen readers see all	Need `aria-setsize` / `aria-posinset`
< a few hundred rows	Thousands of rows, feeds, logs

Perf bug	Fix
Component defined inside component	Move to module level
Unstable / random keys	Stable IDs
Context value recreated each render	Memoize or split context
State too high	Colocate / move down
Effect-driven render loops	Compute in render / handler

Order of fixes: structure first (colocate, children, keys) → memoization → specialised tools.

Code to remember

```
const Row = memo(function Row({ item, onSelect }) { /* ... */ });
function List({ items }) {
  const [sel, setSel] = useState(null);
  const onSelect = useCallback(id => setSel(id), []);   // stable for memo
  return items.map(it => <Row key={it.id} item={it} onSelect={onSelect} />);
}
```

In the interview

Q: When does a component re-render? **A:** Own state, parent re-render, context change, external store change; props only matter with `memo`.

Q: Why can `memo` make things slower? **A:** It compares props every parent render; if they always differ you pay for comparison *and* render.

Q: Does `memo` stop context updates? **A:** No.

Q: What does React Compiler do? **A:** Auto-memoizes components and Hooks at build time based on static dependency analysis; manual memo Hooks become escape hatches. It doesn't fix slow algorithms, big DOM or bundle size.

Q: Why doesn't `memo` fix a 10,000-row list? **A:** The cost is DOM node count and initial render, not repeated re-renders — virtualize.

Q: How would you speed up a slow React page? **A:** Measure (prod build, throttling, Profiler); load time → bundle analysis + `lazy`; interactions → structural fixes, then `memo`/Compiler; virtualize long lists; memoize expensive calcs; transitions for non-urgent updates.

Q: Why must `lazy` be declared at module level? **A:** Inside a component each render creates a new type → remount and lost state.

Gotchas

- `memo` + inline objects/functions/JSX props → always re-renders; `<MemoCard><p>hi</p></MemoCard>` → new `children` each render.
- Deep-equality `arePropsEqual` on big data can cost more than the render.
- `useCallback(() => save(draft), [])` → stale closure; unstable deps (`opts` new each render) → cache never hits.
- `useMemo` isn't a guaranteed cache and doesn't speed up the first render; its function must be pure.
- Splitting everything → many requests and waterfalls.
- Virtual list without fixed viewport height never scrolls; no overscan → blank flashes.
- Treating every re-render as a bug — a 0.1 ms render isn't worth optimising.

TL;DR

Measure, then fix structure (colocate state, children, stable keys, stable context) before `memo`; use the Compiler, split code by route, virtualize long lists.

11. TypeScript with React

Setup, typing props, children, events, state, reducers, refs, Hooks, Context, generic components, discriminated-union props, reading type errors.

Must-know

- **Setup** — `.tsx`, `@types/react` + `@types/react-dom` (matching React major), `strict: true`, `jsx: "react-jsx"`. The bundler **strips** types; `tsc --noEmit` **checks** them (run in CI).
- **Props** — `function C({ a, b = 1 }: Props)`; `?` optional; defaults remove `undefined`; literal unions for variants; function types for callbacks.
- **Children** — `ReactNode` (anything renderable) — not `JSX.Element`/`ReactElement` (a single element only). `PropsWithChildren<P>` helper.
- **Extend native elements** — `ComponentProps<'button'> & { variant: … }`, spread the rest (includes `ref` in React 19).
- **Events** — inline handlers are inferred; standalone: `ChangeEvent<HTMLInputElement>`, `SubmitEvent<HTMLFormElement>` (older: `FormEvent`), `MouseEvent`, `KeyboardEvent` from `react`; or `XxxEventHandler<…>`. `currentTarget` has your element's type.
- **State** — let `useState` infer; add `<T>` for `null`, `[]` and literal unions (`useState([])` alone is `never[]`).
- **Reducers** — actions as a **discriminated union**; `default: const x: never = action` makes missing cases compile errors.
- **Refs** — `useRef<HTMLInputElement>(null)` for DOM; **argument required** in React 19 types; `ref` is a normal prop typed `Ref<T>`; `forwardRef` not needed.
- **Hooks & Context** — tuples need `as const` (or return an object); `createContext<T | null>(null)` + a `useX()` that throws outside the provider.
- **Generic components** — `function List<T>(props: ListProps<T>)`; T inferred from props; constrain with `extends`, `keyof T` for field names; `<T,>` for arrow functions in `.tsx`.
- **Union props** — one member per valid prop combination with a shared literal discriminant; narrow before reading; `?: never` forbids a prop.

Key facts & numbers

Type	Meaning / use
`ReactNode`	Anything renderable → `children`
`ReactElement` / `JSX.Element`	Only a JSX element
`ComponentProps<'input'>` / `ComponentProps<typeof X>`	Reuse native / existing props
`Ref<T>`	`ref` prop type in React 19
`unknown`	Safe alternative to `any` — narrow before use

Code to remember

```ts
type Action = { type: 'add'; sku: string } | { type: 'clear' };
function reducer(state: string[], action: Action): string[] {
  switch (action.type) {
    case 'add': return [...state, action.sku];
    case 'clear': return [];
    default: { const unreachable: never = action; return unreachable; }
  }
}
type ButtonProps = ComponentProps<'button'> & { variant?: 'primary' | 'ghost' };
```

In the interview

Q: How do you type props and children? **A:** Type alias on the destructured parameter, `children: ReactNode`, native elements via `ComponentProps<'button'>`.

Q: `ReactNode` vs `ReactElement` vs `JSX.Element`? **A:** `ReactNode` is anything renderable (strings, arrays, null…); the other two are just a JSX element.

Q: How do you type `ref` in React 19? **A:** A normal prop `ref?: Ref<HTMLInputElement>`, or free from `ComponentProps<'input'>`.

Q: Why avoid `any`? **A:** It disables checking for that value and everything it flows into; use `unknown` and narrow.

Q: How do you prevent invalid prop combinations? **A:** Discriminated unions — invalid combos don't compile (`<Status status="success" />` without `count` errors).

Q: How do you read a TS error? **A:** "Type A (given) is not assignable to type B (expected)" — read the last indented line first.

Gotchas

- Turning `strict` off → lose null checks.
- `children: JSX.Element` → a string or two children fail.
- Spreading `{...rest}` after your own props lets the caller's `className` overwrite yours — merge deliberately.
- `useRef<HTMLDivElement>(null)` on an `<input>` → type error; reading `ref.current` during render.
- `as` and `!` silence the checker without checking; narrow instead.
- Hand-typing API responses and trusting them → validate at the boundary (Zod) and derive types.
- Generics for a single-type component → over-engineering.

TL;DR

`strict` TS, `ReactNode` children, `ComponentProps` for native props, discriminated unions for reducers and variant props, `useRef<T>(null)`, no `any`.

12. Testing React

Testing Library philosophy, queries and priority, `user-event`, async tests, mocking `fetch`/MSW, testing Hooks, providers and custom render, Vitest/Jest setup.

Must-know

- **Philosophy** — test what users see and do: render → find like a user → act like a user → assert on the DOM. Don't test state, props or internal calls.
- **Basic test** — `render` → `screen.getByRole(...)` → interact → assert; `cleanup` automatic in Jest/Vitest.

- **Query priority** — role (+ accessible name) → label → placeholder → text → display value → alt → title → **test ID last**.
- `getBy` / `queryBy` / `findBy` — throws if missing (must exist now) / returns `null` (assert absence) / promise that retries (async appearance). `All` variants for many; `within` to scope.
- `user-event` — `const user = userEvent.setup(); await user.click/type/selectOptions/keyboard/tab(...)`; prefer over `fireEvent`.
- **Async** — `await findBy…` for appearing, `waitForElementToBeRemoved` for disappearing, `waitFor(() => assertion)` for the rest. **Never sleep.**
- **Network** — mock at the boundary (MSW, or a `fetch` spy returning real `Response`s); cover success, HTTP error (`res.ok` false), network error, races.
- **Hooks** — test through a component, or `renderHook` + `act` + `result.current` (in `@testing-library/react` since RTL 13.1); `rerender` for new args; fake timers for time.
- `act` — flushes React updates and effects before assertions; RTL helpers already wrap it — you need it for direct updates (`renderHook` calls, advancing fake timers).
- **Providers** — one `renderWithProviders` using a `wrapper`: `MemoryRouter` with `initialEntries`, context, a **fresh** store/query client per test.

Key facts & numbers

Setup	Config
Vitest	`environment: 'jsdom'`, `globals: true`, setup file importing jest-dom
Jest	`jest-environment-jsdom`, Babel/TS transform, `setupFilesAfterEnv`
Fake timers + user-event	`userEvent.setup({ advanceTimers: vi.advanceTimersByTime })`

Code to remember

```
test('adds a todo', async () => {
  const user = userEvent.setup();
  render(<TodoApp />);
  await user.type(screen.getByRole('textbox', { name: /new todo/i }), 'Milk');
  await user.click(screen.getByRole('button', { name: 'Add' }));
  expect(screen.getByRole('listitem')).toHaveTextContent('Milk');
});
```

In the interview

Q: How do you test React components? **A:** RTL + Vitest/Jest; find by role/name, interact with user-event, assert on the DOM; await async UI; mock the network at the boundary.

Q: Why prefer `getByRole`? **A:** Matches how users and assistive tech find elements, doubles as an accessibility check, survives markup refactors.

Q: How do you test a component that fetches? **A:** Mock with MSW or a `fetch` spy, `await findBy…` the loaded UI, also test loading and error states.

Q: When do you need `act`? **A:** Only for updates outside RTL helpers (e.g. `renderHook` calls, fake timer advances).

Q: How do you test a debounced Hook? **A:** `renderHook` with fake timers, `act(() => vi.advanceTimersByTime(300))`, read `result.current` again.

Gotchas

- `container.querySelector('.btn')` → `getByRole('button', { name })`.
- `getByText` to assert absence throws → `queryByText(...) === null`.

- `getBy` for later elements → `findBy`; don't wrap `findBy` in `waitFor`.
- Actions inside `waitFor` → only assertions there (it may run many times).
- Missing `await` on `findBy` / `waitFor` / `user.*` → passes by accident + `act` warning.
- Destructuring `result.current` once → stale; re-read after each `act`.
- Leaving fake timers on → reset in `afterEach`.
- Importing `renderHook` from `@testing-library/react-hooks` (React ≤17 only).

TL;DR

Test behaviour like a user: role queries, user-event, `findBy` for async, network mocked at the boundary, fresh providers per test; few snapshots, E2E for critical paths.

13. Server Rendering and Frameworks

Rendering strategies, server APIs, streaming SSR, hydration and mismatches, SSG and PPR, Server Components, directives, Next.js App Router, React Router framework mode, choosing a framework.

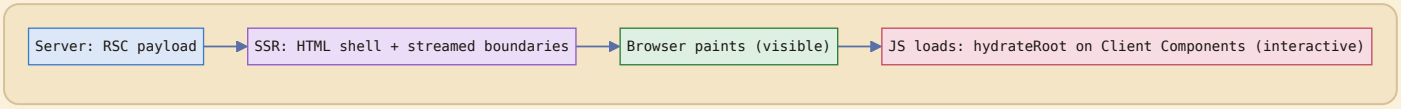
Must-know

- **Strategies** — CSR (browser builds it), SSR (server per request), SSG (at build), **ISR** (SSG + background refresh), **streaming** (frame now, slow parts later), **PPR** (static shell + streamed dynamic holes). Chosen per route, with PPR per component.
- **Server APIs** — `renderToPipeableStream` (Node) / `renderToReadableStream` (Web Streams) for SSR; `prerender*` for build; `resume*` for PPR; `hydrateRoot` in the browser. `renderToString` is legacy (no server Suspense — shows fallbacks).
- **Streaming SSR** — the **shell** (outside pending boundaries) is sent first with fallbacks; each Suspense boundary follows as a hidden div + swap script (`$RC`). Status in `onShellReady`, crawlers wait for `onAllReady`, log in `onError`, `abort()` on timeout.
- **Hydration** — `hydrateRoot` renders the same tree with the same data, adopts the server DOM, attaches handlers/Effects. **Selective**: the shell is interactive while boundaries load; clicks on an unhydrated boundary make React hydrate it first and replay the event. **Visible ≠ interactive.**
- **Hydration mismatch** — first client render ≠ server HTML → React client-renders the tree and reports a recoverable error. Fix by switching after mount: Effect, `useSyncExternalStore` server snapshot, or `use(browser())` (19.3). `suppressHydrationWarning` keeps server text, silences one level deep.
- **Server Components (RSC)** — run only on the server, can be `async`, read data directly, ship **zero JS**; output sent as an **RSC payload**. No directive — they're the default. Share a query across them with `cache()`.
- **Client Components** — `'use client'` marks a **module-graph boundary** (file + its imports ship to the browser); still pre-rendered by SSR, then hydrated.
- **Server Functions** — `'use server'`; async functions callable from the client, for **mutations** (not reads); each is a **public POST endpoint** → auth + validate. Use `import 'server-only'` for private helpers.
- **RSC vs SSR** — SSR makes first-load HTML; RSC decides which components exist in the browser at all. Together: RSC payload → SSR HTML → hydrate only Client Components.
- **Next.js App Router** — folders are routes; `page` / `layout` / `loading` / `error` files; Server Components by default; revalidate with `revalidatePath` / `revalidateTag` / `updateTag`. **Next.js 16**: `proxy.js`, Turbopack default, opt-in **Cache Components** with `'use cache'` + PPR; `params` / `searchParams` / `cookies()` are Promises (since 15; sync removed in 16).
- **React Router framework mode** — `routes.ts` + route modules (`loader`, `action`, `meta`, `ErrorBoundary`, default component); config picks SSR, SPA or prerender; Vite-based.

Key facts & numbers

	CSR	SSR	SSG	ISR	Streaming / PPR
First content	Slow	Medium	Fastest	Fastest	Fast (shell first)
Server needed	No	Yes	No	Yes	Yes
Per-user HTML	No	Yes	No	No	Yes (holes)
Good for	Dashboards behind login	Personalised pages	Docs, blogs	Catalogues, news	Fast frame + slow widgets

Need	Choose
Public, SEO, fast first paint	Next.js or React Router framework mode
Internal tool behind login	Vite SPA + React Router + TanStack Query
Mobile	Expo
Mostly static content	SSG / Astro



In the interview

Q: How does streaming SSR work? **A:** Shell first with fallbacks; each boundary's HTML later with a swap script, so one slow query doesn't block the page and hydration can start early.

Q: What causes hydration mismatches and how do you fix them? **A:** Environment checks, `localStorage`, dates, randomness, locale, invalid nesting, different data. Make the first client render identical; switch after mount.

Q: Why is `useId` SSR-safe? **A:** IDs derive from tree position, identical on server and client.

Q: Server Components vs Client Components? **A:** Server: async, data access, zero JS. Client: state, events, browser APIs. Default to server, push `'use client'` to the leaves, pass server output as `children`.

Q: Can a Client Component render a Server Component? **A:** Not by importing it — only if it receives it as `children`/a JSX prop from a Server Component.

Q: Does `'use client'` mean no server rendering? **A:** No — still SSR'd to HTML; it means the code ships and hydrates.

Q: What does a framework add over Vite + `react-dom/server`? **A:** Routing tied to data and code splitting, data serialization for hydration (no double fetch), caching/revalidation, bundling for server + client, deployment adapters.

Q: Component throws inside a boundary on the server? **A:** Boundary marked for client rendering, fallback stays, browser retries; if it fails there too, the nearest error boundary shows.

Gotchas

- `createRoot` on server HTML → discards and rebuilds the DOM (flash); use `hydrateRoot`.
- `hydrateRoot` on `renderToStaticMarkup` output → no hydration markers.
- `window`/`localStorage` during render → `ReferenceError` on the server.
- One giant `<Suspense>` → empty shell.

- Per-user data in SSG/prerender → everyone sees the first user's cart.
- `'use server'` doesn't mark Server Components; `onClick` from Server to Client Component isn't serializable.
- `useState` in a component without `'use client'` above → build error.
- Server Components re-run only on a new server request (navigation, `router.refresh()`, revalidation).
- `'use client'` on `layout`/`page` or every file → lose RSC benefits.
- SSR for an internal dashboard; SPA for an e-commerce catalogue.

TL;DR

Stream a shell, hydrate selectively, keep the first client render identical to the server. Server Components by default, `'use client'` at the leaves, Server Functions are public endpoints.

14. Patterns and Architecture

Composition, container/presentational, custom Hook extraction, compound components, render props, HOCs, portals, error boundaries, accessibility, folder structure, design systems.

Must-know

- **Composition first** — `children` and slots beat growing lists of boolean props; specialise by wrapping.
- **Container / presentational** — props-in/markup-out views; data access in a Hook, container or Server Component — where reuse or testing benefits, not by rule.
- **Custom Hook extraction** — logic → `useX(inputs)` returning a small, intention-revealing API (`{ status, results }`); injectable deps; test with `renderHook`.
- **Compound components** — `Tabs`, `Tabs.List`, `Tabs.Tab`, `Tabs.Panel` share state via a **private context**; root owns state (controlled + uncontrolled); parts throw a clear error outside the root.
- **Render props** — a function prop the component calls to render; best when the component decides placement/repetition of caller markup (lists, tables, slots). Hooks for plain logic.
- **HOC** — `withX(Component)` returns a wrapped component (e.g. Redux `connect`, auth guards). Apply at **module level**, pass props through, set `displayName`. Mostly legacy — prefer Hooks.
- **Portal** — `createPortal(children, node)` renders DOM elsewhere (escapes `overflow: hidden`, stacking contexts) but the **React tree is unchanged**: context and **event bubbling follow React ancestors**.
- **Error boundary** — class with `getDerivedStateFromError` (fallback) + `componentDidCatch` (log). Catches render-time errors below it — including `use()` rejections and lazy-load failures. **Not**: event handlers, async callbacks, SSR, its own errors. Without one, React 19 unmounts the whole root. `react-error-boundary` adds `resetKeys` and `useErrorBoundary`/`showBoundary`. Root options: `onCaughtError` / `onUncaughtError`.
- **Accessibility** — semantic elements first; `<label htmlFor>` + `useId`; ARIA from state (`aria-expanded={open}`); live regions (`role="status"` polite / `role="alert"` urgent) present before text changes; focus management with refs; `eslint-plugin-jsx-a11y`, axe.
- **Folder structure** — small: by type; growing: **by feature** with colocated tests/styles, a public `index.js` per feature, `app → features → shared` dependency direction enforced by lint; path aliases; careful with giant barrel files.
- **Design system** — tokens → accessible primitives → documented patterns; closed variants, safe defaults, pass-through props and `ref`; build on headless primitives (Radix, React Aria); Storybook.

Key facts & numbers

Reuse pattern	Best for	Downside
Custom Hook	Sharing stateful logic	Doesn't share state
Compound components	Flexible layout of a widget family	Harder to enforce structure
Render props	Component controls where caller markup goes	Callback pyramids
HOC	Cross-cutting wrapping (auth, logging)	Prop collisions, wrapper hell, hidden deps
Config/array prop	Simple fixed cases	New props for every variation

Error boundary catches	Doesn't catch
Render errors in subtree, `use()` rejections, `lazy` load failures	Event handlers, async callbacks, SSR, its own errors

Code to remember

```
const TabsCtx = createContext(null);
function useTabs() { const c = useContext(TabsCtx); if (!c) throw new Error('Tabs.* must be inside
<Tabs>'); return c; }
function Tabs({ value, defaultValue, onValueChange, children }) {
  const [inner, setInner] = useState(defaultValue);
  const selected = value ?? inner;                        // controlled wins
  const select = v => { if (value === undefined) setInner(v); onValueChange?.(v); };
  return <TabsCtx value={{ selected, select }}>{children}</TabsCtx>;
}
```

In the interview

Q: Why context rather than `cloneElement` for compound components? **A:** `cloneElement` only reaches direct children and breaks with wrappers; it's a legacy API. Context works at any depth.

Q: Controlled + uncontrolled support? **A:** Internal state for `defaultValue`; prefer `value` when defined; always call `onValueChange`.

Q: HOC vs Hook? **A:** HOCs compose but cause prop collisions, implicit props, wrapper nesting; Hooks are explicit. HOC still fits when wrapping rendering (guards).

Q: How do events behave in a portal? **A:** They bubble through React ancestors, not DOM ancestors.

Q: Why are error boundaries still classes? **A:** No Hook for `getDerivedStateFromError` / `componentDidCatch`; teams use `react-error-boundary`.

Q: How do you send an async error to a boundary? **A:** Store it in state and throw during render, or `showBoundary(error)`.

Q: How do you make a modal accessible? **A:** Move focus in, trap it (or `inert` background), Escape closes, `aria-modal`, return focus to the trigger — or native `<dialog>` with `showModal`, or Radix/React Aria.

Q: How do you structure a large React app? **A:** By feature with public APIs, `app → features → shared`, lint-enforced boundaries, thin route files.

Gotchas

- `FooContainer` + `Foo` for everything → twice the files, little gain; a "presentational" component calling `useContext` isn't reusable.
- A Hook that uses no Hooks → plain function, don't name it `use…`.
- Hooks inside a render-prop callback → breaks the Rules of Hooks.

- HOC applied inside render → remount every render.
- Error boundary fallback that throws; "Try again" without changing input → use reset keys; always log.
- `<div onClick>` → `<button>`; `outline: none` without `:focus-visible`; ARIA overuse; conditionally rendering the live region itself.
- Portals during SSR → `document` doesn't exist; render after mount.
- `shared/` / `utils/` dumping ground; circular feature imports.
- Design system components that hard-code colours or drop `...rest` / `ref`.

Must-know — client architecture patterns (§14.13)

Pattern	Middle piece	View updated by	Test without UI
MVC	Controller handles input → Model	view observes the model	controller + model
MVP	Presenter formats data	presenter calls a **passive** view interface	presenter with a fake view
MVVM	ViewModel: observable state + commands, no view reference	view **binds** to state	ViewModel alone
VIPER	Interactor (use cases) + Presenter + Router (navigation), Entities	presenter	every part
Flux / Redux / MVI	pure reducer	action → reducer → state → view (one-way)	reducer as a pure function

- **React** — a view library with one-way flow, `UI = f(state)`; a **custom Hook ≈ ViewModel**, `useReducer`/Redux = unidirectional, stores via `useSyncExternalStore`.

Q: MVP vs MVVM? **A:** MVP's presenter holds the view and calls it; MVVM's ViewModel only exposes state and the view binds to it.

Q: Is React MVC? **A:** No — it is the view layer; Hooks play the ViewModel role and reducers/stores give Flux-style one-way updates.

TL;DR

Compose first, extract Hooks for logic, compound components via context for widget families; boundaries per route; accessibility is semantic HTML + focus + ARIA from state; organise by feature.

15. React Interview and Machine Coding

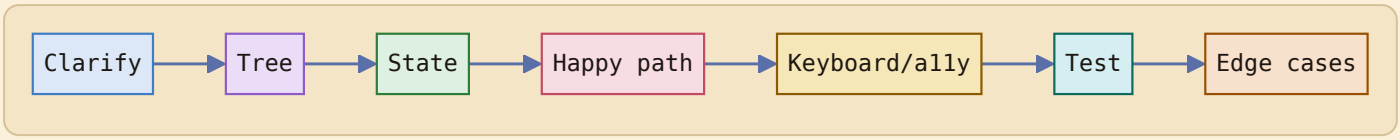
How React interviews run, the machine-coding method, the 53 questions (condensed), and the 8 builds as requirements → state design → key code idea → edge cases.

Must-know

- **Rounds** — concepts ("what does this log?"), machine coding (45–90 min widget build), JavaScript (closures, event loop, `debounce` from scratch), front-end system design (feed, chat, design system).
- **Machine-coding method** — 1 clarify requirements (incl. quiet ones: keyboard, screen readers, loading/error, data size) → 2 sketch component tree → 3 design state (derivable? who needs it? outside React?) → 4 happy path with native elements → 5 keyboard + a11y → 6 one or two user-like tests → 7 edge cases. **Talk throughout** — interviewers score process.
- **State design rules** — minimal state, derive the rest; colocate, lift only when shared; one `status` value, not several booleans; refs for values handlers need but the screen doesn't.

Key facts & numbers

60-minute budget	Minutes
Requirements + tree + state	10
Happy path	25
Keyboard + a11y	8
Test	7
Edge cases, discussion	10



In the interview

The 53 questions of §15.2–15.9, condensed (ones already answered in Ch 1–14 above are merged there; these are the extras and the "what does this log" facts):

Q: What happens when you call a state setter? **A:** Nothing immediately: the update is queued → render (call components, reconcile) → commit (DOM diff, layout effects), browser paints, then `useEffect` runs.

Q: Why is reconciliation O(n) and not O(n³)? **A:** Two heuristics — a different element type means a different subtree; keys identify list children. Identity = **type + position + key**. "Virtual DOM" is the older name for the in-memory element tree.

Q: What does changing `key` do? **A:** Different instance: old unmounts (state gone, Effects cleaned up), new mounts fresh — `<Profile key={userId} />`.

Q: What is batching; did React 18 change it? **A:** All updates in one event → one render. Before 18 only React handlers were batched; with `createRoot` it's automatic everywhere (timeouts, promises, native listeners). `flushSync` forces a synchronous update.

Q: What's wrong with `useState(prop)`? **A:** It uses only the prop's first value. Use the prop directly, name it `initialX`, or reset with `key`.

Q: React events vs DOM events? **A:** Synthetic event wrapper; since React 17 listeners sit on the **root container** (not `document`); propagate through the React tree; `onChange` fires every keystroke; returning `false` doesn't prevent default.

Q: Effect order? **A:** Children's Effects before the parent's; all layout effects before passive ones; cleanup runs before re-run with the **old** values, and on unmount.

Q: What does StrictMode double? **A:** Dev only: component bodies, initialisers, updaters run twice; Effects setup → cleanup → setup; since React 19 ref callbacks too. Fix cleanups; don't add "has run" flags.

Q: `useRef` vs `useState` precisely? **A:** Both persist; state re-renders and is a snapshot; `ref.current` changes immediately and doesn't re-render.

Q: Is Context a state manager? **A:** No, a transport; state lives in `useState`/`useReducer` in a provider. React 19 renders `<ThemeContext value>` directly.

Q: What can suspend? **A:** `lazy()` components, `use(pending promise)`, Suspense-enabled sources (`useSuspenseQuery`, framework loaders, async Server Components) — **not** data fetched in `useEffect`.

Q: Does the fallback reappear when on-screen content re-suspends? **A:** Not inside a transition — old content stays (dim it with `isPending`); without a transition the nearest fallback shows. Routers wrap navigations in transitions.

Q: `'use client'` on the root layout? **A:** Everything becomes client code — bigger bundle, no zero-JS parts, no direct data access. Push it to leaves.

Q: Error thrown in an `onClick` — which boundary shows it? **A:** None; boundaries catch rendering, lifecycle and Effect errors. Catch in the handler and put the error in state (or throw it inside `startTransition` / via a rejected `use(promise)`).

Q: Memoized child, no props, reads context with `value={{ theme }}`; parent re-renders — does the child re-render? **A:** Yes: new object → context changed → context updates bypass `memo`.

Q: Why is `key={Math.random()}` worse than index? **A:** Index only misbehaves on reorder; random remounts every item every render.

Q: Can a Server Component pass `onClick={() => …}` to a Client Component? **A:** No — not serializable. Pass data or a `'use server'` Server Function.

Build 1 — Todo list with filters

- **Requirements:** add (Enter/button, ignore blank), toggle, delete, filter All/Active/Completed, edit (double-click; Enter saves, Escape cancels, empty deletes). Quiet: keyboard-only, accessible names, persist to `localStorage`, announced counter ("2 items left"), toggle-all, clear completed.
- **State design:** `todos` via `useReducer` in `TodoApp` (many kinds of change → one pure reducer); `filter` and new-todo `text` state in `TodoApp`; `editing`/`draft` **colocated** in each `TodoItem`; `visible` and `left` **derived**; next ID in a ref.
- **Key code idea:** event-named actions (`added`, `toggled`) in a pure, unit-testable reducer; lazy initialiser `useReducer(reducer, storageKey, loadTodos)` reads storage once, an Effect writes back; ID keys so filtering never moves the edit box; a `cancelled` ref so Escape beats the blur-save; filter buttons with `aria-pressed`.
- **Edge cases:** server persistence with `useOptimistic` or a TanStack mutation; filter in the URL; two tabs → `storage` event dispatches `replaced`; 10,000 todos → virtualize + `memo(TodoItem)`; undo via a `past` stack; `crypto.randomUUID()` for real IDs.

Build 2 — Autocomplete (debounce, abort, keyboard)

- **Requirements:** suggestions as you type, click/Enter selects, ArrowUp/Down with wrap, Escape closes, loading/empty/error states, highlighted match. Quiet: **debounce 300 ms**, ignore **stale responses**, ARIA **combobox** pattern. Data source is a prop `search(query, { signal })`.
- **State design:** `query` (urgent, controlled); `debounced` inside `useDebouncedValue`; `result = { status, items, for }` as one object; `open`; `active` index (−1 = none); `showList` derived; option IDs from `useId`.
- **Key code idea:** debounce the *value* in an Effect (cleanup clears the timeout); a search Effect keyed on `debounced` owns an `AbortController` aborted in cleanup (ignore `AbortError`); focus stays in the input and `aria-activedescendant` names the active option; `onMouseDown={e => e.preventDefault()}` on options stops blur from closing the list before the click; Enter selects only with a highlighted option and `preventDefault` so a surrounding form doesn't submit; `Highlight` splits the string and renders `<mark>` in JSX (XSS-safe).
- **Edge cases:** cache query → results in a ref `Map` or TanStack Query; debounce vs throttle (pause vs at-most-once per interval); cap results or `useDeferredValue`; `scrollIntoView({ block: 'nearest' })` for the active option; rate limits → debounce + `minChars` + caching.

Build 3 — Infinite scroll

- **Requirements:** first page, load the next near the bottom, "Loading more…", end message, error + Retry. Quiet: **never two requests for the same page**, keep loading until the box is full, no state updates after unmount, works inside a scroll box.
- **State design:** `items`, `page`, `status` (`'idle' | 'loading' | 'error'`), `hasMore` as state; `inFlight` **ref** (synchronous guard — state changes only next render); `mounted` ref; refs to the scroller and

sentinel.

- **Key code idea:** an `IntersectionObserver` on a 1-px sentinel after the last item, `root` = scroll box, `rootMargin: '200px'` to prefetch. The Effect observes only while `status === 'idle'` and `hasMore`; after each page deps change, cleanup disconnects, and the new observer fires at once if the sentinel is still visible — so short pages chain. Retry resets `status` to `'idle'`. `aria-busy` + `role="status"`.
- **Edge cases:** `useInfiniteQuery` + observer calling `fetchNextPage()`; virtualize thousands; **cursor** (`?after=<lastId>`) over page numbers when inserts shift pages, de-dupe by `id`; restore scroll on back-navigation; offer a "Load more" button for keyboard users; abort on unmount.

Build 4 — Modal dialog (portal, focus trap, Escape)

- **Requirements:** open from a button; title, content, actions; close by Cancel, Escape, backdrop click; destructive "type the name to enable Delete". Quiet: rendered at top level (no clipping), focus **in** on open and **back** to the opener on close, Tab/Shift+Tab cycle inside, background doesn't scroll, `role="dialog"` + `aria-modal` + `aria-labelledby`.
- **State design: controlled** — parent owns `open` and passes `onClose`; the modal holds only refs (dialog node, previously focused element).
- **Key code idea:** `createPortal(…, document.body)` (React tree unchanged, so events bubble to React parents); an Effect on `open` saves `document.activeElement`, focuses the first focusable element, locks body `overflow`, and its **cleanup** restores focus and overflow; keydown traps Tab (last → first, Shift+Tab first → last, re-querying focusables each press); Escape closes with `stopPropagation` for nested modals; backdrop closes only when the **mousedown target is the backdrop** (text selection dragging out won't close it).
- **Edge cases:** prefer native `<dialog>` + `showModal()` in production (top layer, inert background, Escape via `cancel`, focus return); `inert` on the app root; nested modal stack; exit animation via a `closing` state; SSR → portal after mount; long-tail focusables → `focus-trap`/Radix/React Aria.

Build 5 — Accessible tabs

- **Requirements:** click a tab to show its panel, optional initial tab + `onChange`, skip disabled tabs, keep typed input in other panels. Quiet (WAI-ARIA tabs): `tablist`/`tab` (`aria-selected`, `aria-controls`)/`tabpanel` (`aria-labelledby`), **one Tab stop** for the row, ArrowLeft/Right wrap, Home/End, automatic activation.
- **State design:** one state — `selected` index; IDs from `useId`; `tabRefs` array filled by ref callbacks; tab list derived from the `tabs` prop.
- **Key code idea: roving tabindex** — current tab `tabIndex={0}`, others `-1`; `nextEnabled` walks with wrap-around skipping disabled tabs (Home/End start just outside the ends), then `.focus()` via `tabRefs`; `preventDefault` so Home/End don't scroll; panels use **`hidden`** instead of conditional rendering so inactive panels stay mounted and keep state.
- **Edge cases:** manual activation (separate `focused` index, Enter/Space selects); controlled version (`?tab=billing`); compound API via context; disabled-but-focusable with `aria-disabled`; heavy panels → `<Activity mode="hidden">` (19.2); vertical tabs with `aria-orientation`.

Build 6 — Star rating

- **Requirements:** 5 stars, click to rate, "3 of 5", hover preview, clear, controlled, configurable `max`. Quiet: Tab to the group + arrows, announced group name / star labels / checked state, read-only mode.
- **State design:** `value` is a **prop** (parent owns it); `hover` is the only state (visual preview); `shown = hover ?? value` derived; group `name` from `useId`.
- **Key code idea:** a rating is **one of N → a radio group**: real `<input type="radio">` visually hidden inside `<label>`s in a `<fieldset>` + `<legend>` — one Tab stop, arrows, checked state and screen-reader output for free; `checked={value === n}`, `onChange={() => onChange(n)}`; `onMouseLeave` on the fieldset resets hover; Clear calls `onChange(0)`.

- **Edge cases:** half stars (10 radios, clipped halves; read-only average as a percent-width overlay); visible focus via `input:focus-visible + .star`; uncontrolled version with `defaultValue` + `name` for `FormData`; no hover on touch; `role="radiogroup"` + `aria-labelledby` as the ARIA alternative.

Build 7 — Pagination

- **Requirements:** 10 rows/page, Previous/Next, numbered buttons, rows-per-page (5/10/25), "21–25 of 47". Quiet: `1 … 4 5 6 … 20` with first/last always visible, **constant button count**, `<nav aria-label="Pagination">`, `aria-current="page"`, disabled ends, never an empty page when data shrinks.
- **State design:** `page` and `pageSize` state in the table; `totalPages = max(1, ceil(n / pageSize))`, rows slice and range text **derived**; `current = min(page, totalPages)` **clamped** in render — no Effect to "fix" state.
- **Key code idea:** split into a **pure** `getPageRange(current, total, siblings)` (with `siblings = 1` → 7 slots; `…` only replaces two or more hidden pages; near the ends show the first/last five) tested with a table of cases, plus a **controlled** `<Pagination page totalPages onPageChange>`; changing page size resets to page 1.
- **Edge cases:** server pagination keyed `['orders', page, pageSize]` with `keepPreviousData`; page in the URL; offset (jump anywhere, slow deep, shifts on insert) vs cursor (stable, next/prev only); clamping keeps the stored page → reset to 1 on filter change; move focus to the table heading after a page change; huge counts → no total or an estimate.

Build 8 — Multi-step form wizard with validation

- **Requirements:** Account → Profile → Review → Create account; validate per step on Next; Back keeps input; step indicator; pending + success/error on submit. Quiet: `aria-invalid` + `aria-describedby`, focus the first invalid field, `aria-current="step"`, focus the new step heading, no double submit, retry after failure.
- **State design:** one `data` object for **all** steps; `step`; `errors` `{ field: message }` (set on Next, cleared as the field is edited); `status` `'editing' | 'submitting' | 'done' | 'failed'`; `STEPS` config constant outside the component with fields and a **pure** `validate(data)` per step.
- **Key code idea:** one `<form noValidate>` so Enter triggers Next/Create; on errors focus the first invalid field; `goTo` updates state inside `flushSync` then focuses the `<h2 tabIndex={-1}>` ("Step 2 of 3: Profile") — in the handler, not an Effect, so it doesn't fire on mount; both buttons disabled while submitting.
- **Edge cases:** React 19 `<form action>` + `useActionState` + `useFormStatus` (Actions reset only uncontrolled fields); React Hook Form + Zod per step (`trigger([...])`); validate again on the server; save progress in `sessionStorage` (never the password) or step in the URL; conditional steps derived from `data`; prefix IDs with `useId`; async "email taken" check debounced on blur/Next.

Other prompts — core state → technique

Prompt	Core state	Technique
Accordion	`openIds` Set or one ID	`<button aria-expanded aria-controls>` or `<details>`
Dropdown / menu	`open`, `active`	Autocomplete keys + Modal Escape/click-outside + focus return
Toasts	queue `{ id, text }`	Portal, `role="status"`, timers cleared in cleanup
Tooltip / popover	`open`	Portal, `useLayoutEffect` to position, `aria-describedby`, Popover API
Countdown / stopwatch	`startedAt`, `elapsed`	Interval in Effect; compute from timestamps, not tick counts
Carousel	`index`	Wrap-around; pause on hover/focus; `aria-roledescription="carousel"`
Nested tree	`expanded` set	Recursive components; ARIA `tree` + arrows
Data table	`sort`, `filters`, `page`	Derive (`useMemo` if large), Pagination, URL state
Kanban	columns → card IDs	Normalised reducer; `@dnd-kit`; keyboard alternative
OTP input	digits array	Refs array for focus; paste; `autoComplete="one-time-code"`
Chat	messages + per-message status	`useOptimistic`, auto-scroll ref, virtualize history

Gotchas

- Coding before agreeing requirements; silence while coding; no time left for edge cases.
- Storing `filteredTodos` next to `todos` → derive it.
- `<div onClick>` / `<span onClick>` stars → native `<button>` / radios.
- Several booleans (`isLoading`, `isError`) → one `status`.
- Using state as a synchronous guard against double requests → use a ref.

TL;DR

Clarify → tree → minimal state → happy path → keyboard/a11y → test → edge cases, out loud. Each build has one trick: reducer + derived filter, debounced value + abort + activedescendant, sentinel observer + ref guard, portal + focus trap/restore, roving tabindex, native radios, pure range + clamp, per-step pure validators + focus.

16. Cheat Sheets and Glossary

Every Hook in one table, the common-errors table and the silent bugs — the night-before sheet.

Must-know

- All Hooks follow the Rules of Hooks — except `use`, which may be called in `if`/loops (not in `try/catch`).
- React 19 added `use`, `useActionState`, `useFormStatus` (`react-dom`), `useOptimistic`; 19.2 added `useEffectEvent` and `<Activity>`.
- Related non-Hook APIs: `memo`, `lazy` + `<Suspense>`, standalone `startTransition`, `createContext` / `<Ctx value>`, `createPortal`, `flushSync`, `<Profiler>`, `<StrictMode>`, `<Activity>`, `cache()` (Server Components).

- Lifecycle order: trigger → render (pure) → commit → layout effects → paint → Effects; children's Effects before parents'.

Key facts & numbers

Hook	For	Gotcha	Since
`useState`	Memory that triggers re-render	Snapshot; lazy init with a function; `Object.is`-equal → bail-out	16.8
`useReducer`	Related updates in a pure reducer	Pure (run twice in StrictMode); `dispatch` stable	16.8
`useContext`	Nearest provider's value	Default only with **no** provider; re-renders through `memo`	16.8
`use`	Read a promise (suspends) or context	Promise must be cached; allowed in `if`/loops	19.0
`useRef`	Non-rendering values, DOM nodes	No re-render; not during render; `null` until commit	16.8
`useImperativeHandle`	Expose a limited API	Pair with `ref` prop	16.8
`useEffect`	Sync with external system after commit	Not `async`; not on server	16.8
`useLayoutEffect`	Measure/fix before paint	Blocks paint; nothing on SSR	16.8
`useInsertionEffect`	CSS-in-JS `<style>` injection	Not for app code	18.0
`useEffectEvent`	Latest values inside an Effect	Only call from Effects; never in deps	19.2
`useMemo` / `useCallback`	Cache value / function identity	A cache, not a guarantee; missing deps → stale	16.8
`useTransition`	Non-urgent updates, Actions	Not for input value; updates after `await` need own `startTransition`	18.0
`useDeferredValue`	Lagging copy for a slow child	Needs `memo`; not a debounce; `initialValue` in 19	18.0
`useId`	SSR-safe `htmlFor`/`aria-*` IDs	Never for keys	18.0
`useSyncExternalStore`	Subscribe to outside stores, no tearing	Cached snapshot; stable `subscribe`	18.0
`useDebugValue`	Label a custom Hook in DevTools	Library Hooks	16.8
`useActionState`	Last Action result + `isPending`	`fn(prevState, payload)`; queued	19.0
`useFormStatus`	Parent form status	In a child; from `react-dom`	19.0
`useOptimistic`	Expected result during Action	Call inside Action; auto rollback	19.0

Error message (dev)	Usual cause → fix
Too many re-renders	Setter called during render (`onClick={setOpen(true)}`) → pass a function
Objects are not valid as a React child	Rendering an object/Date/Promise → render a field
Each child in a list should have a unique "key"	Missing key / key on wrong element → `key={item.id}` on outermost mapped element
Encountered two children with the same key	Duplicate IDs → truly unique stable ID
The tag <profile> is unrecognized	Lowercase component → capitalise
Invalid DOM property `class`	HTML attribute names → `className`, `htmlFor`
`value` *prop … without an* `onChange`	Controlled without handler → `onChange`, `defaultValue` or `readOnly`
Changing an uncontrolled input to be controlled	`value` undefined then string → init `''`
Rendered more/fewer hooks	Conditional Hook / early return → Hooks unconditionally at top
Invalid hook call	Hook outside component, or two copies of React → `npm ls react`
getSnapshot should be cached → Maximum update depth	New object each snapshot/selector → same reference, `useShallow`
Suspended by an uncached promise	`use(fetch(...))` in render → cache/loader/`useSuspenseQuery`
Lazy element type must resolve to a class or function	No default export → `.then(m => ({ default: m.X }))`
Hydration failed … text didn't match	Non-deterministic render → same first render, switch after mount
Not wrapped in act(...)	Missing `await` in test → `findBy`/`waitFor`
Cannot assign to read only property	Mutating props (frozen in dev) → callback to parent
TS2322 `never` / TS18047 null / TS2554 `useRef`	`useState<string[]>([])` / narrow / `useRef<T>(null)`

Silent bug symptom	Cause → fix
+1 instead of +3	Snapshot → updater functions
Screen doesn't update	Mutation → new object
Typed text jumps rows	Index keys → stable IDs
Input loses focus each keystroke	Nested component definition / random keys
Old search results win	Fetch race → abort/ignore flag
`memo` child still re-renders	Inline props or context → stabilise/split
Interval sees first value	Stale closure → updater / `useEffectEvent`
Prop change doesn't reset child	Same type + position → `key`
Click in modal triggers parent's "click outside"	Portal events bubble through React → check `e.target`
404 on refresh in SPA	No `index.html` fallback on the server

In the interview

Q: Which Hook can be called conditionally? **A:** `use` only (not inside `try/catch`).

Q: What did React 19 add? **A:** Actions with `useActionState`, `useFormStatus`, `useOptimistic`, `use()`, `ref` as a prop, ref-callback cleanups, `<Ctx value>` as provider; 19.2 `useEffectEvent`, `<Activity>`; 19.3 Fragment refs and `use(browser())`.

Gotchas

- `useFormStatus` import is `react-dom`, not `react`.
- `useDeferredValue` and `useTransition` are not debounces.
- `useId` output format is not a stable API — never parse it.

TL;DR

One table of Hooks, one table of errors, one table of silent bugs — read the message, map it to the cause.

Last-minute checklist

- [ ] Trigger → render (pure) → commit → layout effects → paint → Effects; children's Effects first.
- [ ] Identity = type + position + key; new `key` = fresh instance; never define components inside components.
- [ ] Keys: stable IDs; index only for static lists; never `Math.random()`.
- [ ] `&&` with a number renders `0`; objects can't be children.
- [ ] State is a snapshot; updaters chain; batching everywhere since 18; `flushSync` opts out.
- [ ] Never mutate; spread / `map` / `filter` / `toSorted` / `with`; Immer for deep.
- [ ] Minimal state, derive the rest; one `status`, not several booleans.
- [ ] Effects only for external systems; cleanup mirrors setup; no `async` setup; StrictMode double run is dev-only.
- [ ] Fetch in Effect → `AbortController` in cleanup + `res.ok`; production → TanStack Query / loaders.
- [ ] Refs don't re-render and aren't read in render; `ref` is a prop in 19; `useImperativeHandle` for limited APIs.
- [ ] Rules of Hooks: top level, React functions only; `use` is the exception.
- [ ] Context re-renders all consumers on a new value, through `memo` → memoise/split or use a selector store.
- [ ] TanStack Query: key includes every variable; `staleTime` 0, `gcTime` 5 min, 3 retries; `isPending` vs `isFetching`.
- [ ] Forms: `<form action>`, `useActionState(prev, formData)`, `useFormStatus` in a child from `react-dom`, `useOptimistic` inside an Action; validate on the server.
- [ ] Performance: measure → structure → `memo`/Compiler → `lazy` at module level → virtualize.
- [ ] `useTransition` wraps updates, `useDeferredValue` wraps values; neither debounces.
- [ ] SSR: shell streams first, selective hydration, visible ≠ interactive; mismatches from non-deterministic render.
- [ ] RSC: server by default, `'use client'` at leaves, only serializable props, `'use server'` = public endpoint.

- [ ] Error boundaries: class, render/Effect errors only; portals bubble through the React tree.
- [ ] Testing: `getByRole` + name, `user-event`, `await findBy`, never sleep, mock the network.
- [ ] Accessibility: native elements, `htmlFor` + `useId`, ARIA from state, focus management, live regions.
- [ ] Machine coding: clarify → tree → state → happy path → keyboard/a11y → test → edge cases — out loud.